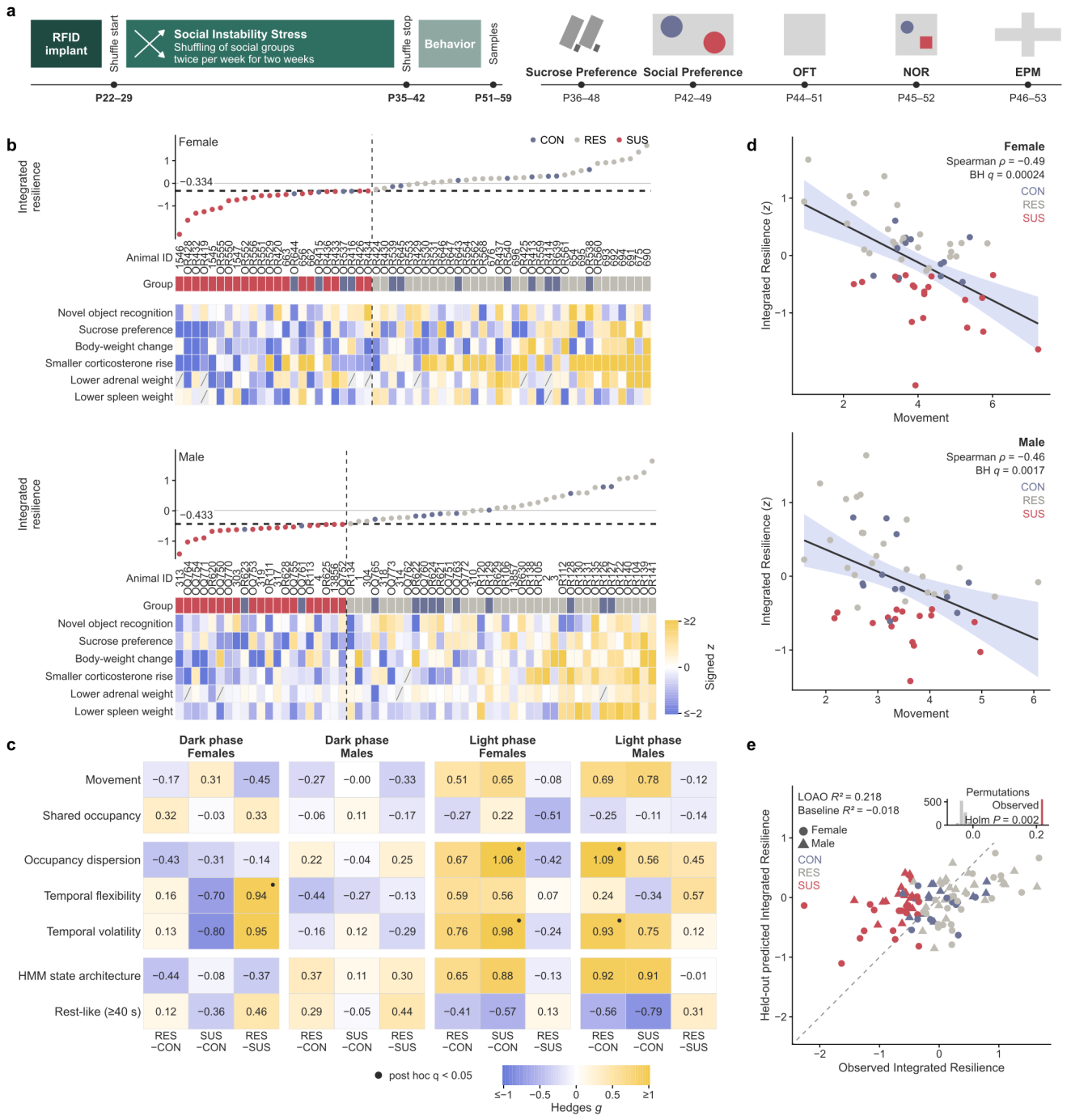

a
RFID implant
Shuffle start
Social Instability Stress
Shuffling of social groups
twice per week for two weeks
Shuffle stop
Behavior
Samples
P22–29
P35–42
P51–59
Sucrose Preference
Social Preference
OFT
NOR
EPM
P36–48
P42–49
P44–51
P45–52
P46–53
b
Female
CON
RES
SUS
Integrated resilience
1
0
−1
−0.334
Animal ID
Group
Novel object recognition
Sucrose preference
Body-weight change
Smaller corticosterone rise
Lower adrenal weight
Lower spleen weight
Male
−0.433
≥2
0
≤−2
Signed z
c
Dark phase Females
Dark phase Males
Light phase Females
Light phase Males
Movement −0.17 0.31 −0.45 −0.27 −0.00 −0.33 0.51 0.65 −0.08 0.69 0.78 −0.12
Shared occupancy 0.32 −0.03 0.33 −0.06 0.11 −0.17 −0.27 0.22 −0.51 −0.25 −0.11 −0.14
Occupancy dispersion −0.43 −0.31 −0.14 0.22 −0.04 0.25 0.67 1.06 −0.42 1.09 0.56 0.45
Temporal flexibility 0.16 −0.70 0.94 −0.44 −0.27 −0.13 0.59 0.56 0.07 0.24 −0.34 0.57
Temporal volatility 0.13 −0.80 0.95 −0.16 0.12 −0.29 0.76 0.98 −0.24 0.93 0.75 0.12
HMM state architecture −0.44 −0.08 −0.37 0.37 0.11 0.30 0.65 0.88 −0.13 0.92 0.91 −0.01
Rest-like (≥40 s) 0.12 −0.36 0.46 0.29 −0.05 0.44 −0.41 −0.57 0.13 −0.56 −0.79 0.31
RES −CON
SUS −CON
RES −SUS
post hoc q < 0.05
≤−1 −0.5 0 0.5 ≥1
Hedges g
d
Female
Spearman ρ = −0.49
BH q = 0.00024
CON
RES
SUS
Integrated Resilience (z)
Movement
Male
Spearman ρ = −0.46
BH q = 0.0017
e
LOAO R² = 0.218
Baseline R² = −0.018
Female
Male
Permutations
Observed
Holm P = 0.002
Held-out predicted Integrated Resilience
Observed Integrated Resilience